

PASCO TRI-CITIES, WA, USA

WMO: 727845

Lat: 46.267N

Lon: 119.117W

Elev: 407

StdP: 14.48

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24163

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.2	17.1	3.2	6.6	14.4	9.7	9.2	20.3	28.7	48.3	25.6	46.6	4.1	330	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.5	99.2	69.1	96.3	67.7	92.7	66.5	71.4	93.4	69.4	91.6	67.8	88.8	6.8	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.9	90.3	79.2	61.9	84.0	76.7	59.8	77.9	75.0	35.3	93.5	33.6	91.6	32.1	88.7	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	20.5	18.0	DB	3.3	105.3	9.3	3.2	-3.4	107.6	-8.9	109.5	-14.1	111.3	-20.9	113.6	(4)
(5)			WB	2.7	75.6	9.2	3.6	-4.0	78.1	-9.4	80.2	-14.5	82.3	-21.2	84.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	35.0	38.6	46.1	53.1	61.4	68.1	75.4	73.7	64.4	52.3	41.3	33.9	(6)	
	DBStd	15.90	8.87	7.87	6.77	5.69	6.79	5.94	5.52	5.36	5.96	6.90	8.71	8.71	(7)	
	HDD50	1811	466	323	152	31	3	0	0	0	1	53	281	501	(8)	
	HDD65	4952	929	739	586	359	153	32	2	4	80	395	710	963	(9)	
	CDD50	3164	2	5	31	124	356	543	788	734	433	125	20	3	(10)	
	CDD65	830	0	0	0	2	41	125	325	272	63	2	0	0	(11)	
	CDH74	12999	0	0	3	95	826	1960	4943	3937	1181	54	0	0	(12)	
	CDH80	6529	0	0	0	16	303	858	2816	2086	443	7	0	0	(13)	
(14) Wind	WSAvg	6.6	6.0	6.5	7.6	7.9	7.4	7.7	6.3	5.9	5.5	5.9	6.1	5.8	(14)	
(15) Precipitation	PrecAvg	7.8	1.1	0.7	0.7	0.6	0.6	0.4	0.2	0.4	0.3	0.6	1.0	1.1	(15)	
	PrecMax	13.7	3.9	1.8	1.7	1.9	2.3	1.2	1.5	1.7	1.6	1.9	2.9	3.5	(16)	
	PrecMin	3.8	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	(17)	
	PrecStd	2.1	0.7	0.5	0.4	0.4	0.5	0.3	0.3	0.4	0.4	0.5	0.7	0.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.7	62.5	72.2	82.0	93.0	99.1	104.9	101.5	93.4	80.6	67.8	59.6	(19)	
		MCWB	50.3	51.8	54.9	60.7	65.4	70.1	71.1	68.6	67.4	61.0	58.0	51.4	(20)	
	2%	DB	54.3	57.4	66.2	76.2	88.0	93.3	100.2	98.0	89.3	73.4	62.7	54.6	(21)	
		MCWB	47.0	48.2	51.8	57.7	63.9	66.8	69.3	68.5	65.0	58.2	52.4	47.9	(22)	
	5%	DB	50.7	54.4	62.8	71.8	82.6	89.1	97.5	94.7	85.2	69.7	58.1	50.3	(23)	
		MCWB	44.7	46.2	50.3	55.2	61.2	64.8	68.4	67.0	63.3	55.7	50.0	44.7	(24)	
	10%	DB	47.2	51.4	59.3	67.1	78.5	84.4	94.0	91.1	81.2	65.8	54.4	46.4	(25)	
		MCWB	42.5	44.2	48.0	52.0	59.2	62.4	66.9	65.7	61.7	53.8	47.6	41.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	53.0	56.7	63.1	70.0	72.1	76.8	72.9	69.0	62.8	59.3	52.1	(27)	
		MCDB	57.4	60.9	67.9	79.2	85.2	93.8	97.1	94.6	88.9	75.9	66.5	58.9	(28)	
	2%	WB	47.7	49.8	53.5	59.4	65.6	68.8	72.3	70.6	66.6	59.7	53.6	48.3	(29)	
		MCDB	53.2	56.3	63.4	73.5	82.7	89.9	94.6	93.0	85.2	71.6	60.3	53.8	(30)	
	5%	WB	45.3	46.8	51.3	56.4	62.9	66.3	70.0	68.7	64.6	57.3	51.1	45.5	(31)	
		MCDB	50.2	53.2	60.3	69.5	80.0	85.5	92.9	90.4	81.7	67.0	56.4	49.9	(32)	
	10%	WB	42.7	44.7	49.4	53.5	60.5	63.9	68.1	67.0	62.6	55.1	48.7	42.2	(33)	
		MCDB	46.9	50.1	58.2	65.1	75.1	81.4	90.2	87.3	78.7	63.9	53.9	46.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	18.5	23.6	26.1	28.0	28.5	33.5	32.8	31.3	24.8	17.0	12.4	(35)	
		MCDBR	18.0	21.8	28.7	34.3	36.6	36.7	39.5	38.4	37.7	29.2	20.4	17.2	(36)	
	5% WB	MCWBR	12.4	14.0	16.3	17.5	15.4	13.9	13.7	13.5	15.6	15.3	12.7	12.1	(37)	
		MCWBR	17.4	20.3	26.1	32.9	32.1	34.6	37.4	36.2	35.1	26.7	18.9	17.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.290	0.292	0.306	0.345	0.362	0.360	0.363	0.359	0.343	0.313	0.297	0.285	(40)	
		taud	2.557	2.569	2.546	2.403	2.378	2.405	2.400	2.405	2.440	2.531	2.540	2.547	(41)	
	Ebn at Noon	Ebn at Noon	263	285	295	289	287	287	284	281	277	272	256	251	(42)	
		Edh at Noon	20	23	28	35	37	36	36	34	30	24	20	18	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	373	716	1132	1677	2033	2284	2438	2046	1515	938	488	308	(44)	
	RadStd	RadStd	46	92	79	97	126	111	80	111	85	70	43	36	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.21	N/A	N/A	N/A	+259	+126	(46)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	N/A	(47)

Nomenclature: See separate page